

PABLO UBILLA PAVEZ

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EDUCATION

PhD	Machine Learning — INRIA & Université de Montpellier
2025–Now	Thesis on <i>Deep Learning Methods for Species Distribution Models</i> . Supervised by Diego Marcos & Christophe Botella.
MSc	Ecology and Data Science — University College London
2023–2024	Degree awarded with Highest Honours (Top 5%). Thesis on <i>Consensus Clustering for Functional Groups</i> . Supervised by Daniel Maynard.
Master	Operations Management — Universidad de Chile
2022–2023	Degree awarded with Highest Honours (Ranked 1/15). Thesis on <i>Outlier Detection for Elections with Demographic Groups</i> . Supervised by Charles Thraves.
Prof. Degree & BSc	Industrial Engineering — Universidad de Chile
2017–2023	Degree awarded with Highest Honours (Ranked 1/174). Exchange period at Universiteit Leiden (Spring 2022).

RESEARCH INTERESTS

Machine Learning, Statistics, and Ecology.

PUBLICATIONS

2026 Ubilla Pavez, P., Hermosilla, D. & Thraves, C. *An accurate, fast, and scalable ecological inference algorithm for the $R \times C$ case*. Statistics and Computing, 36, 195 (2026). [doi:0.1371/journal.pcbi.1014278](https://doi.org/10.1371/journal.pcbi.1014278)

2026 Ubilla Pavez, P., Paz, A., & Maynard, D. S. *Functional group classification using consensus clustering*. PLOS Computational Biology, 22(5), 1–25. [doi:10.1371/journal.pcbi.1014278](https://doi.org/10.1371/journal.pcbi.1014278)

SOFTWARE

faitei: Ecological inference methods (R package in CRAN, 2025). Work with Thraves, C., and Hermosilla, D.

WORK EXPERIENCE

Oct 2024 – Sep 2025	Airnguru , <i>Data Scientist</i>
Remote, Chile	Developed automation of large-scale pricing strategies (up to billions of fares).
Jan 2021 – Jan 2022	Complex Engineering Systems Institute , <i>Research Assistant</i>
Santiago, Chile	Adaptive question selection using dynamic programming, greedy algorithms, and reinforcement learning to maximize the information obtained at the end of a survey.
Sep 2022 – Jan 2023	Municipalidad de Vitacura , <i>Data Scientist</i>
Santiago, Chile	Estimated daily street-level recycling collection and optimized truck routes using GPS historical data, leading to a redesigned partition of municipal collection sectors.
Aug – Oct 2020	Fundación Tierra Viva , <i>Intern</i>
Santiago, Chile	Contributed in data collection for <i>Resumen del Estado del Medio Ambiente 2020</i> .

TEACHING EXPERIENCE

*Seasons correspond to the hemisphere of the institution. †TA: Teaching Assistant; EA: Exam Assistant; I: Instructor.

FRB-CESAB

June 2026 **Introduction to Deep SDMs (I)**, AI for Ecologists: a toolkit

University College London

Fall 2023 **Financial Engineering (TA)**, MSc Computational Finance
Spring 2024 **Applied Computational Finance (TA)**, MSc Computational Finance

Universidad de Chile

Fall 2023 **Modelling and Optimization (TA)**, Industrial Engineering
Fall 2020 **Modelling and Optimization (TA)**, Industrial Engineering
Spring 2022 **Decisions Under Uncertainty (TA)**, Industrial Engineering

Spring 2021	Decisions Under Uncertainty (TA), Industrial Engineering
Spring 2021	Game Theory and Strategy (TA), Industrial Engineering
Fall 2021	Probabilities (TA), Industrial Engineering
Fall 2021	Operations Research (TA), Industrial Engineering
Spring 2020	Operations Research (TA), Industrial Engineering
Fall 2022	Probabilities (EA), Industrial Engineering
Fall 2022	Information Engineering (EA), Industrial Engineering
Spring 2019	Modelling and Optimization (EA), Industrial Engineering
Fall 2020	Introduction to Calculus (EA), Mathematics Engineering
Spring 2019	Introduction to Algebra (EA), Mathematics Engineering
Fall 2019	Probabilities and Statistics (EA), Mathematics Engineering

Universidad Finis Terrae

Fall 2023	Stochastic Models (I), Industrial and Informatics Engineering
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Universidad Adolfo Ibáñez

Summer 2022	Operations Research (TA), Industrial Engineering
Spring 2021	Operations Research (TA), Industrial Engineering

AWARDS & SCHOLARSHIPS

2024	Best Presentation Prize , MSc Ecology and Data Science UCL
2024	Dean's List , MSc Ecology and Data Science UCL
2024	Distinguished Master's Graduate Award , Postgraduate Engineering School UCH
2023	Masters Abroad Scholarship , ANID (Chile), Full scholarship (tuition and living stipend)
2022	Academic Excellence Scholarship , MSc Operations Management
2022	International Mobility Scholarship , Santander Bank
2020	Best Teaching Assistant , Industrial Engineering UCH
2018–2021	Outstanding Student , Faculty of Physics and Mathematics UCH

TALKS & POSTERS

Deep Learning Integration for Species Distribution Models

2026	World Biodiversity Forum (<i>Davos, Switzerland</i>) (Talk)
2026	DeepSDM Day (<i>Sion, Switzerland</i>) (Poster)
2026	The European Conference for Biodiversity Monitoring (<i>Montpellier, France</i>) (Talk)
2026	Rencontre CiSStats-RESSTE (<i>Avignon, France</i>) (Talk)

Consensus Clustering for Functional Groups

2026	INRIA PhD Seminar (<i>Montpellier, France</i>) (Talk)
2024	British Ecological Society: Macroecology (<i>Cardiff, UK</i>) (Talk)

A Fast Ecological Inference Algorithm for the RxC case

2025	Workshop in Management Science (<i>Pucón, Chile</i>) (Poster)
2024	Operational Research Society 66 (<i>Bangor, UK</i>) (Talk)

Optimized sensor deployment in biodiversity monitoring

2024	Surrey Wildlife Trust Showcase (<i>Surrey, UK</i>) (Talk)
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DEVELOPMENT

Summer 2026	Probabilistic AI School — Artificial Intelligence Association of Lithuania
Vilnius, Lithuania	Probabilistic modelling; variational inference; deep generative models.
Summer 2026	Probability meets Biology III — University of Bath
Bath, UK	Collaborative 4 day workshop between biologists and mathematicians.
Summer 2023	XVIII Summer School in Discrete Mathematics — ISCV
Valparaíso, Chile	High-chromatic graphs; provable algorithms for machine learning; linear programming.

SKILLS

Programming	Python, R, SQL, MATLAB	Machine Learning	TensorFlow, PyTorch, Scikit-Learn
Optimization	Gurobi, OR-Tools, CVXPY	Other	Git, L ^A T _E X, HPC